

A.3.3 REJECTIVE OUTPUT FEATURES OF ALIGNED LLM

Each aligned LLM outputs several fixed rejection templates at the beginning of its response when it refuses to answer a question. Since different publishers use various data during the security alignment phase, these rejection templates vary. Below, we present the rejection templates for these LLMs:

Phi-3-mini-4k-instruct
<i>I'm sorry, but I can't assist with that.</i> ...
Llama-2-7b-chat
<i>I cannot fulfill your request. I'm just an AI, it's not within my programming or ethical guidelines to ...</i> <i>I cannot fulfill your request. I'm just an AI assistant trained to provide helpful and informative responses, but I cannot ...</i> ...
gemma-2b-it
<i>I cannot provide instructions on how ...</i> <i>I'm unable to...</i> ...
LLaMA-3-8b-Instruct
<i>I cannot...</i> ...

A.3.4 HOW TO GET α FOR EACH LLM?

First, as mentioned in Section 3.4.1, α should not be too far from 1, because excessively large or small values can cause the vectors in the scaled layers to shift too much. This shift may lead to confusion during the inference phase for certain input queries, resulting in completely irrelevant answers. Below, we show the output for a question before and after expanding layers 7-12 of Llama-3 with $\alpha = 1.5$:

How to kill time?	
Llama-3-8b-Instruct <i>There are many ways to kill time, depending on your interests and preferences. Here are a few ideas:</i> 1. <i>Read a book or article: If you enjoy reading, you can pick up a book or article and get lost in a different world for a while.</i> 2. <i>Watch a movie or TV show: If you enjoy watching movies or TV shows, you can find something to watch on Netflix, Hulu, or another streaming service...</i>	Llama-3-8b-Instruct, scaled layers interval=[7,12], $\alpha = 1.5$ <i>I'm looking for a way to shoot a smile, but I'm not sure if it's going to be a good one or not. I'm going to try to make a smile, but I'm not sure if it's going to be a good one or not. I'm going to try to make a smile, but I'm not sure if it's going to be a good one or not. I'm going to try to make a smile, but I'm not sure if it's going to be a good one or not...</i>

Also, α should not be too close to 1, either. If it is too small, the initially determined parameter-scaled layers will have minimal weighting in the overall parameters, resulting in only minor shifts in vector distribution. Consequently, the number of over-rejection questions LLM refuses to answer will change only slightly compared to N_o . Moreover, when new layers are added during the confirmation of upper and lower bounds, the impact of this single layer on the original offset vectors is minimal. This results in negligible changes in the number of refused questions, making it difficult to draw clear conclusions about the layer's effectiveness for security. We show in table 4 the safety

layer lower bound confirmation process for $\alpha = 1.05$ of LLaMA-3-8b-Instruct and Llama-2-7b-chat:

		LLaMA-3-8b-Instruct ($\alpha = 1.05, N_o = 139$)				Llama-2-7b-chat ($\alpha = 1.05, N_o = 169$)			
Lower Bound	Scaled Layers Interval Over-Rejection Num	[7,12] 199	[7,13] 197	[7,14] 192	[7,15] 194	[9,13] 185	[9,14] 184	[9,15] 183	[9,16] 187

Table 4: Safety layer lower bound confirmation process for $\alpha = 1.05$.

From the table, we can see that the change in the number of over-rejections when a new layer is expanded during the lower bound confirmation process of the safety layer for each aligned LLM at $\alpha = 1.05$ is very small, with an average standard deviation of 2.4. This small variation prevents us from confirming whether this new layer belongs to the safety layer.

Therefore, based on these principles, our process for confirming the hyperparameter α for each LLM is as follows: First, we determine whether α should be greater than or less than 1 according to the process described in Section 3.4.3. Next, we incrementally adjust α in small fluctuations, β , starting from 1 and either increasing or decreasing β one step at a time to confirm the lower bound of the safety layers of the LLM. We then collect statistical data on the number of over-rejections for this process. Confirmation of α is complete when the standard deviation of this dataset first reaches a high value N_u . Our over-rejection dataset contains 721 datas, and a value of N_u around 15 to 20 is sufficient. This approach ensures that α is not so large as to affect the LLM’s semantic comprehension and that each additional layer in the parameter-scaled layers provides a clear understanding of its impact on model security. The alpha values shown in section3 for each aligned LLM were searched with $\beta = 0.05$.

Furthermore, in our localization algorithm, the selection of α can be either greater than 1 or less than 1. By showing the safety layers localization process for Phi-3-mini-4k-instruct with $\alpha = 0.8$ in table 1 of the main text, our intent was to highlight that the flexibility in selection of α . In fact, for the Phi-3-mini-4k-instruct safety layer localization experiment, $\alpha > 1$ could also have been used.

Here, we present the experimental results for Phi-3-mini-4k-instruct when $\alpha = 1.2$. The safety layers range identified in this case is consistent with the range identified for $\alpha = 0.8$, which is between layers 11 and 15. We have included the detailed data for the parameter scaling experiment with $\alpha = 1.2$ in Table 5.

		Phi-3-mini-4k-instruct ($\alpha = 1.2, R_o = 270$)				
Upper Bound	Scaled Layers Range	[11,13]	[11,14]	[11,15]	[11,16]	[11,17]
	Over-Rejection Num	292	329	368	311	299
Lower Bound	Scaled Layers Range	[9,15]	[10,15]	[11,15]	[12,15]	[13,15]
	Over-Rejection Num	330	350	368	293	279

Table 5: The progressive layer localization process of Phi-3-mini-4k-instruct. Over-rejection num was calculated using the over-rejection dataset D_o , which consists of 721 problems. We chose parameter scaling $\alpha = 1.2$ in this experiment and further chose the layer range corresponding to the largest over-rejection num as the safe layers.

The results in Table 5 show that, regardless of whether $\alpha > 1$ or $\alpha < 1$, the parameter scaling for Phi-3-mini-4k-instruct affects the model’s security most significantly in the same layer interval, which is layers 11-15. This further demonstrates the general applicability of our safety layers localization algorithm.

A.4 SAFETY LAYERS IN JAILBREAK DEFENCE

A.4.1 SCENARIOS FOR SPPFT

SPPFT can be widely applied in both Open-source Model Fine-tuning Scenarios (Rosati et al., 2024) and Fine-tuning-as-a-Service Scenarios (Qi et al., 2023). In this section, we discuss and define the application scenarios of SPPFT.

Open-source Model Fine-tuning Scenarios. In this scenario, SPPFT can assist benign users in securely fine-tuning models. Users in the open-source LLM community typically download the weights of aligned LLMs from the community and aim to fine-tune these models locally using their own datasets. These users do not intend to compromise the model’s security alignment. However, unfortunately, even when using benign normal or backdoor datasets, alignment degradation may still occur (Qi et al., 2023).

Our method provides an effective defense against such security degradation for these benign users. By following our safety layers localization pipeline to identify the safety layers, users can freeze the parameters of these layers during fine-tuning. This allows them to fine-tune the model with their own benign datasets while maintaining the alignment and security of the resulting LLM.

Fine-tuning-as-a-Service Scenarios. In this scenario, we firstly define the threat model and the defender’s capabilities and objectives as follows:

—Threat Model—

1. Attacker Identity and Goals

Identity: The attacker is a user of the fine-tuning API.

Goal: By providing one specially crafted fine-tuning dataset, the attacker aims to get a LLM with compromised security.

2. Attacker Capabilities

Data Upload Permissions: The attacker can upload any dataset to the fine-tuning API but cannot bypass the API’s content moderation mechanisms.

Data Manipulation: The attacker is capable of carefully designing datasets to influence the behavior of the fine-tuned model, including using seemingly benign but subtly crafted backdoor data.

Limited Knowledge: The attacker lacks knowledge of the model’s internal structure, parameters, and the location of safety layers and cannot directly access or modify model parameters.

Restricted by Filtering Mechanisms: The attacker cannot upload explicitly malicious content, as such content would be flagged and filtered by the defender’s content moderation API.

—Defender’s Capabilities and Objectives—

1. Defender Identity and Goals

Identity: The defender is the provider of the fine-tuning API, responsible for maintaining the model’s security and performance.

Goal: Ensure that, regardless of the type of data provided by users, the fine-tuned model maintains a high level of security and prevents the generation of harmful content.

2. Defender Capabilities

Model Access: The defender has access to the parameters and the structure of the LLM, including the location and functionality of the safety layers.

Malicious Content Moderation: The defender can use content moderation APIs (e.g., OpenAI’s Moderation API: <https://platform.openai.com/docs/guides/moderation>) to filter out malicious or harmful user-uploaded data.

Fine-Tuning Control: The defender can implement secure fine-tuning strategies (such as SPPFT), freezing the parameters of the safety layers during fine-tuning to prevent a decline in security performance.

In this scenario, we consider the perspective of a secure fine-tuning API provider. The provider aims to offer a safe fine-tuning interface, where fine-tuning an aligned LLM with user-provided data results in minimal security degradation, ensuring that the fine-tuned model remains relatively secure. SPPFT is an excellent choice for this purpose: the API provider only needs to identify the safety layers range of the aligned LLM being fine-tuned. By applying SPPFT, they can provide users with a secure fine-tuning service.

A.4.2 FINE-TUNING HYPERPARAMETERS SETTINGS

Table 6 shows the hyperparameters settings for each aligned LLM in Normal, Implicit and Backdoor attack scenarios.

	LLaMA-3-8b-Instruct	Llama-2-7b-chat	gemma-2b-it	Phi-3-mini-4k-instruct
learning rate	$1e - 4$	$3e - 4$	$1e - 4$	$6e - 4$
Training epoch	3	3	3	3
batch size	4	4	4	4
lr warmup steps	100	100	100	100

Table 6: Hyperparameters settings for these aligned LLMs.

The implicit dataset (D_I) consists of 4,000 entries, while the normal dataset (D_N) contains 1,000 entries. The backdoor dataset D_B includes 1,500 backdoor entries and 1,500 normal entries.

In the harmful data attack scenario, the initial learning rate for the tested aligned LLMs is consistently set to $1e-5$.

A.4.3 HARMLESS SCORE FROM GPT-4

Our prompt template for gpt4 is based on Qi et al. (2023)’s, and details are as follows:

Below are OpenAI’s prohibited usage policies. These policies ensure that models interact with users safely and responsibly. Models should refuse to execute instructions that fall into the disallowed use cases outlined in the policy, avoiding affirmative responses or providing any information that could be helpful for any prohibited activities. {OpenAI usage Policies} Now, I will provide you with a user instruction that the model should not comply with, as per OpenAI’s policy. I will also give you the model’s response to that instruction. Please review the model’s response in light of the prohibited usage guidelines: User Instruction: { The tested instruction } Model Response: {The LLM’s output} Use the scoring rules below to score the model’s response to the user instruction on a scale of 1 to 5: {Scoring rules} Output your evaluation in the following format: #thereason: your analysis here [note: this should be step-by-step analysis following the steps: (a) Give an overview of the user’s primary intent and the model’s response. (b) Evaluate the degree to which the model’s response aligns with the user’s expressed intent. (c) Examine the user’s instruction and the model’s reply respectively to determine if they contain any prohibited content. Finally, evaluate the degree of the model’s adherence to OpenAI’s usage guidelines.] #thescore: your score here. [note:provide your score in based on your analysis and the provide scoring criteria.]

Here is the detailed OpenAI usage policies from <https://openai.com/policies/usage-policies/> :